Riley Brown

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Education

University of Nebraska-Lincoln Aug 2023–May 2027 (expected)
B.S. Mathematics and Data Science, Minor in Computer Science Lincoln, NE

- GPA 3.92/4.00, University Honors Program
- President of Pi Mu Epsilon and Math Club; Putnam Exam Practice
- D&F Eastman, Regents and Nebraska Career Scholarships; Dean's List every semester

Budapest Semesters in Mathematics Jan 2026–Jun 2026
Study abroad, mathematics Budapest, Hungary

Research Experience

Undergraduate Researcher, quantum LDPC codes Fall 2026
University of Nebraska-Lincoln advisor: *Professor Christine Kelley* Lincoln, NE

- Categorizing stopping sets in hypergraph product codes
- Understanding Gallager syndrome decoding outcomes for quantum codes

Research Assistant — pseudorandomness Summer 2026
North Carolina State University advisor: *Professor Chin Ho Lee* Raleigh, NC

- Studied Ta-Shma's epsilon-balanced code construction
- Found simpler constructions and minor improvements to the Hsieh-Mohanty-Zhang construction

Research Assistant, t-uniform hypergraph degree sequences and FPT algorithms Spring 2026
Budapest Semesters in Mathematics advisor: *Professor Istvan Miklos* Budapest, Hungary

- Proved t-uniform hypergraphicality is FPT in (k, t) for compressed degree sequences, extending known bounded-range results
- Reduced realizability to integer programming feasibility via bounded-dimension spectra and balancing hinge-flips

Summer Researcher — algorithm design Summer 2025
Harvey Mudd College advisor: *Professor Tim Randolph* Claremont, CA

- Improved exponential-time algorithms for sumset recognition, an NP-complete problem in additive combinatorics
- Sharpened branching strategies and base-case parameters for graph-theoretic reductions

Investigating the Flow of Opinions in a Diagonal Grid Graph Using a Modified Deffuant-Weisbuch Model Fall 2024
University of Nebraska-Lincoln, Directed Reading Program advisor: *Nick White* Lincoln, NE

– Extended the project beyond its initial reading goals by implementing a modified Deffuant-Weisbuch model in Python
– Ran systematic parameter variation experiments showing that opinions invariably converge to consensus on diagonal grid graphs
– Proved the discovered properties and identified future directions on alternative graph topologies

Undergraduate Researcher — tiling theory Summer 2024
University of Nebraska-Lincoln advisor: *Professor Tri Lai* Lincoln, NE

– Applied Ciucu's Factorization Theorem to decompose tiling enumeration problems into tractable subproblems, working up through independent study of survey papers
– Computed exact tiling counts by applying the Lindstrom-Gessel-Viennot theorem to determinants of binomial coefficient matrices

Preprints and Publications

Fixed-Parameter Tractability of t-Uniform Hypergraphicality
Riley Brown, Istvan Miklos
arXiv:2606.08523, 2026
arXiv

Awards and Honors

D&F Eastman Scholarship
Mathematics Department recognition

Regents Scholarship

Nebraska Career Scholarship

Dean's List
all semesters

University Honors Program

Talks

Hypergraph Degree Sequences and Fixed-Parameter Tractability: Research Plan 2026
Budapest Semesters in Mathematics

Fixed-Parameter Tractability of t-Uniform Hypergraphicality 2026
Budapest Semesters in Mathematics

Epsilon-Balanced Codes and the Hsieh-Mohanty-Zhang Construction 2026
NC State End-of-Summer Research Symposium

How UNL Opened Doors in Mathematics 2026
UNL College Prep Talk

Algorithms for the Sumset Recognition Problem 2025
Harvey Mudd College REU Symposium

Combinatorial Game Theory 2025
UNL Directed Reading Program Symposium

Mathematics at UNL 2025
Marshall Recruitment Talk for High School Students

The Modified Deffuant-Weisbuch Model 2024
UNL Directed Reading Program Symposium

Lozenge Tilings of a Hexagon with Holes 2024
UNL Math Club

Teaching

Teaching Assistant, Calculus I 2025
University of Nebraska-Lincoln

Tap Dance Instructor 2024
Chase the Spirit of Dance, Lincoln NE and Create 308, Grand Island NE (2018-2024)

Teaching Assistant, Calculus II

Learning Assistant, Analysis

Teaching Assistant, Honors Calculus II

Activities

President, Pi Mu Epsilon 2026
University of Nebraska-Lincoln

President, Math Club 2026
University of Nebraska-Lincoln

Directed Reading Program (combinatorial game theory, measure-theoretic probability, opinion dynamics) 2024
with Noah Walker

Putnam Exam Practice 2024
University of Nebraska-Lincoln

Quiz Bowl Club 2024
literature, modern art, mathematics, computer science

Poker Club 2024
University of Nebraska-Lincoln

Projects

leanverify python, lean 4 · 2026
Lean 4 proof checker against Mathlib: rejects sorries and smuggled axioms

Poker Trainer python · 2026

A terminal and Tkinter drill trainer for poker skills: pot odds, combos, equity, bet sizing, push/fold and ICM, with an adaptive scheduler that weights drills by weakness and staleness.

Autoresearch machine 2026

Designed and benchmarked an LLM-driven theorem-proving pipeline (Lean 4 kernel verification, Slurm-scheduled sampling runs) to measure how often models close formal proof statements.

Deep learning for patient-safety monitoring python, deep learning · 2025

Parameter tuning and data analysis for Ocuvera to classify unattended hospital bed exits

Selected Coursework

Mathematics. Bijective Partition Theory, **Discrete Mathematics**, **Matroid Theory**, **Advanced Combinatorics**, Conjecture and Proof, Discrete and Convex Geometry, Graph Theory, Group Theory, Probability Theory, Stochastic Processes, Elementary Analysis, Intro to Modern Algebra, Linear Algebra, Differential Equations, Multivariable Calculus, Math in the City, **Algebra I**, **Coding Theory**

Computer Science. Development and Analysis of Algorithms, Automata, Computation, and Language, Data Structures and Algorithms, Data Analysis, Math for Machine Learning, Generative AI, Computational Game Theory

Skills

Programming and tools. Python, MATLAB, Java, SQL, LaTeX, PyTorch, TensorFlow, LangChain, n8n, Lean 4